# WEEK 4 PROJECT SUMMARY REPORT

**Project:** HealthConnect Clinic Experience Lab
**Project Title:** Improving Patient Appointment Attendance and Healthcare Support using Data and AI.

## 1. Introduction

As a Data Scientist working for the HealthConnect Clinic, I analysed the available HealthConnect dataset provided and developed evidence-based recommendations for the features that the Machine Learning team should consider for a future appointment no-show prediction model. My selected ***Target Variable*** was ***Appointment Outcome***.

## 2. Resources Used

The Week 4 assessment used the available appointment dataset, relevant dataset variables, the HealthConnect Data Dictionary, and machine learning concepts related to feature selection, target definition, data quality, and data leakage. The analysis also considered the timing and availability of features before an appointment.

## 3. Key Observations

The initial assessment identified several potentially useful feature groups, particularly historical attendance behaviour, appointment timing, reminder information, accessibility, and booking characteristics.
Historical attendance variables are considered high-value predictors because previous attendance behaviour may provide information about future attendance. Booking and appointment timing information also appears suitable because it is generally available before the appointment.
Reminder variables require additional investigation because their suitability depends on when the reminder information becomes available. Post-appointment outcome information should be excluded because it could introduce target leakage.
Demographic variables should also be reviewed carefully to determine whether they provide legitimate predictive value without creating fairness or ethical concerns.

## 4. Proposed Approach

The proposed modelling approach is to first define a clear binary target representing whether a scheduled appointment results in a no-show. Candidate features will then be selected based on whether they are available at the intended prediction point.
The initial modelling stage will include data cleaning, missing-value assessment, categorical-variable handling, feature selection, and a baseline classification model. Model performance will then be evaluated using appropriate classification metrics, with particular attention to identifying patients at risk of non-attendance.

## 5. Key Considerations

Key considerations include data quality, missing values, class imbalance, feature availability, target leakage, and the timing of information used for prediction. The treatment of cancellations must also be clearly defined so that cancellations are not incorrectly treated as no-shows.
The project should also consider fairness and responsible use of demographic information, as well as ensuring that the model is useful from a healthcare operational perspective.

## 6. Proposed Focus for Week 5

The focus for Week 5 will be data preparation, Feature engineering, and Baseline model development for appointment no-show prediction.